“Business Case”

1.Business Need (Problem or Opportunity): -

Problem Statement

The fresh graduate hiring process is inefficient, costly, and poorly matched. Globally, **45% of graduates are underemployed within their first year**, while recruiters spend around **23 hours per junior role** screening applications.

In **Pakistan**, over **500,000 graduates enter the job market annually**, yet:

- **60–70% fail to find relevant employment within 6 months**
- Around **40% of applications are irrelevant**
- Employers spend **5–10 minutes per CV**

This results in wasted time, increased hiring costs, and poor job-market alignment.

In the **United States**:

- **4.1 million graduates annually**, with **18.5% unemployment (0–12 months)**
- Average **7.2 months to first job**
- **$4,700 cost per hire**
- **11.3 hours wasted per role** due to unqualified applications

Overall, this leads to an estimated **$648 million annual economic loss** from inefficient hiring.

Opportunity:

The **global recruitment software market is valued at $3.2B (2025)**, with an estimated **$1.2B addressable niche in fresh graduate hiring**.

Currently, no major platform is dedicated exclusively to fresh graduates—existing solutions (LinkedIn, Indeed, Handshake) are either broad or internship-focused.

Capturing just **0.5% of this niche** provides strong entry potential, supported by a **14–18 month first-mover advantage** in a growing market segment.

2.Project Justification and Strategy: -

Project Justification

The project addresses the high operational costs and inefficiencies of manual recruitment. By automating the initial screening phase, the platform delivers immediate financial value to stakeholders.

Key Financial Impact (Monthly):

- **Cost Reduction:** Automating screenings reduces manual labor costs by **40%**, lowering the monthly per-company expense from **PKR 20,000** to **PKR 12,000**.
- **Collective Savings:** For the initial 50 target employers, this generates a total saving of **PKR 400,000/month**.
- **Revenue Potential:** Leveraging government incentives (e.g., $2,400 tax credits per hire), the platform aims to capture a **5% service fee** per successful placement.
- **Market Demand:** Validation is confirmed by **327 signed Letters of Intent (LOIs)** from universities, representing a potential user base of **1.2 million students**.
- **Cost of Delay:** Postponing implementation results in an estimated **$48,000 in lost revenue** per month.

Business Strategy

The project follows a **niche-to-mass market strategy**, focusing on entry-level hiring to lead the digital recruitment transformation in the region.

Strategic Roadmap:

- **Immediate Focus:** Dominating the graduate hiring market through AI-driven matching and replacing legacy email-based systems.
- **Long-Term Vision:** Scaling operations across all major universities in Pakistan to become the nation's centralized graduate hiring ecosystem.

Quantified Strategic KPIs:

Strategic Goal	Baseline (Today)	Year 1 Target
Youth Unemployment Reduction	18.5%	12.0% (↓ 6.5%)
University Partnerships	0	50 Institutions
AI Portfolio Growth	1 Product	2 Products (↑ 100%)

3.Proposed Solution: -

The proposed solution is a high-performance, web-based **AI Job Portal** featuring a **Bidirectional Matching Engine**. This platform eliminates manual screening by digitally connecting verified student talent with employer requirements.

Core Features

- **Student Hub:** Profile management and AI-optimized CV uploads.
- **Employer Suite:** Job posting dashboard with automated candidate ranking.

- **Admin Console:** Verification oversight and system-wide analytics.
- **Verification Layer:** Integration with University APIs to validate academic credentials (e.g., verifying a 3.5 CGPA).

AI Matching Logic

The engine transforms raw resume data into actionable hiring scores through a three-step process:

1. **AI Resume Parser:** Extracts **20+ data points** and 10–20 specific keywords per CV (including GPA, technical skills, and project experience).
2. **Matching Algorithm:** Utilizes **NLP techniques** (TF-IDF or Cosine Similarity) to compare candidate profiles against job descriptions.
3. **Quantified Scoring:** Generates a **Match Score (0–100%)** to rank applicants.
    - *Example: Candidate score = 78% → Recommended.*

4.Feasibility Summary: -

The project's feasibility is assessed across five key dimensions to ensure technical viability, economic return, and operational success within the defined constraints.

Multi-Dimensional Analysis

Dimension	Assessment & Quantified Data	Status
Technical	**85% Feasibility.** Development utilizes Python for AI and React/Django for the interface. Open-source tools minimize licensing costs.	**Achievable**
Economic	Total Estimated Cost: **PKR 401,000**. This covers development, hosting, setup, and 6 months of maintenance. **Break-even point: Month 14.**	**Viable**
Operational	Integrates seamlessly into existing "Campus Recruitment" workflows. Adoption target: **1,000 students and 50 companies** in 3 months; <1 hour training time.	**High**
Schedule	Total timeline: **16 weeks (120 days)**. This includes phased planning, design, development (5 weeks), testing, and deployment.	**Realistic**
Legal	**100% Compliance** with Data Privacy Acts. Requires user consent for resume parsing and utilizes secure, encrypted CV storage.	**Compliant**

Resource & Timeline Breakdown

- **Team Composition:** 3–4 developers + 1 Project Manager.
- **Infrastructure Cost (Annual):** PKR 36,000 for hosting + PKR 5,000 for domain/setup.

- **Maintenance:** Allocated PKR 60,000 for the first 6 months post-launch to ensure zero-downtime operations.

5.Expected Benefits and Value: -

Financial Impact & ROI

The platform utilizes a high-margin revenue model designed for rapid recovery of the **PKR 401,000** initial investment.

- **Monthly Revenue: PKR 400,000** (via job postings, premium subscriptions, and ads).
- **Net Monthly Profit: PKR 350,000** (after operational expenses).
- **Payback Period: ~1.15 Months**.
- **First-Year ROI: ~947%**.
- **Long-term Value:** 3-year Net Present Value (NPV) of **$45,000**.

Efficiency & Strategic Gains

Metric	Baseline	Target
Time-to-Hire	42 Days	18 Days (↓ 57%)
Cost Per Hire	Standard	60% Reduction
Matching Quality	Manual/Variable	AI-Data Driven

Qualitative Benefits

- **Precision Placement:** AI-driven alignment of student skills with employer needs.
- **Market Impact:** Directly addresses the youth unemployment gap through accessible entry-level hiring.
- **User Satisfaction:** Modernizes the "Campus Recruitment" experience with transparent, real-time tracking.

6.High-level Cost and Risks: -

Estimated Project Costs

The budget covers initial development through the first year of operations, combining labor and infrastructure.

Category	Initial Investment	Annual/Recurring
Development Team	$20,000 (PKR 300,000 labor)	—
Cloud Infrastructure	$1,200	$1,200/year (PKR 36k)

Marketing & Outreach	$3,500	—
AI API & Maintenance	$800	$800/year (PKR 60k)
Total Investment	**$25,500 (PKR 401,000)**	**PKR 50,000/month**

Risk Assessment & Mitigation

Risk	Probability	Impact	Mitigation Strategy
AI Mismatch	Medium	High	Refine TF-IDF/Cosine Similarity with human-in-the-loop validation.
Low Initial Adoption	High	Medium	Execute targeted marketing and leverage the 327 university LOIs.
Data Breach	Low	High	Implement secure encryption protocols and GDPR compliance audits.
Development Delay	Medium	Medium	Utilize Agile tracking and 2-week sprint cycles to monitor progress.